CLASSGUARD AI

Product Requirements Document

Version: 1.0 **Status:** Draft for validation **Date:** 03 November 2026

Product vision

Beyond Attendance. Towards Accountability.

A cross-device virtual classroom platform for schools, colleges, universities and coaching institutes. This document defines the product scope, users, requirements, architecture direction, privacy principles, MVP and acceptance criteria.

Prepared as the master product blueprint for the initial build.

Document Control

Item	Details
Product	ClassGuard AI
Document	Product Requirements Document (PRD)
Version	1.0
Status	Draft — decisions and assumptions require validation
Primary platform	Responsive web application with PWA support
Future clients	Android, iOS and desktop applications
Target markets	Schools, colleges/universities and coaching institutes
Primary product outcome	Secure, reliable and interactive online teaching

How to use this PRD

Treat this document as the source of truth for product scope and acceptance criteria. Any change to MVP scope, data collection, monitoring behaviour or user permissions should be recorded as a decision before implementation.

Important product principle

A connected device does not prove that a student is attentive. ClassGuard AI will report observable events—attendance duration, tab visibility, responses, quiz participation and connection state—without presenting uncertain signals as facts about a student's mental state.

1. Executive Summary

ClassGuard AI is a multi-tenant SaaS platform for conducting live online classes. It combines video classrooms, verified access, attendance records, classroom moderation, interactive learning activities and teacher/institution analytics in one responsive web application.

The platform addresses common weaknesses in remote teaching: passive or unattended meeting sessions, unreliable attendance, unauthorized participants, classroom disruption and limited visibility into student participation.

Initial strategy: build a browser-first platform that works on laptops, tablets and phones. Use a proven real-time video infrastructure rather than building media servers from scratch. Add AI features only where they provide measurable value and can be explained to users.

2. Product Vision, Goals and Non-Goals

Vision

Make online classes safer, more interactive and easier to manage for teachers and institutions, while respecting student privacy and accessibility.

Product goals

- Enable institutions to create and manage their own private online learning environments.
- Provide reliable live audio/video classes across modern mobile and desktop browsers.
- Make classroom entry controlled by authentication, enrollment and host permissions.

- Record accurate join, leave, reconnect and participation events.
- Give teachers useful, explainable classroom insights and moderation controls.
- Support quizzes, polls, hand-raising, chat and learning resources.
- Keep institutional data isolated and protect student information, including information about minors.

Non-goals for the first release

- Replacing a full learning management system with advanced grading, curriculum authoring and complex admissions.
- Guaranteeing that a student is mentally attentive based on camera, face or head-pose signals.
- Secretly monitoring other applications, personal browsing, messages or device activity.
- Building custom video codecs, global media infrastructure or native mobile apps before validating the web MVP.
- Automated disciplinary decisions or attendance penalties based solely on AI predictions.

3. Target Customers and Personas

Persona	Needs	Primary capabilities
Institution owner / Super Admin	Manage institutions, plans and platform health	Tenant provisioning, subscriptions, support controls, system metrics
Institution Admin	Manage people, courses and policy	User import, role assignment, class setup, reports, settings
Principal / HOD	Monitor department or school operations	Aggregated attendance, class activity and institutional reports
Teacher	Run classes and understand participation	Schedule/start class, admit/remove users, polls, quizzes, attendance, repo
Student	Join classes easily and participate	Timetable, join flow, video controls, chat, hand raise, quizzes, resources
Parent / Guardian	Understand a linked student's progress	Read-only attendance and approved progress information

Customer segments

- **Schools:** supervised access, parent visibility, age-appropriate privacy and simple teacher workflows.
- **Colleges/universities:** departments, large class rosters, attendance exports and course-level analytics.
- **Coaching institutes:** recurring batches, high-volume live sessions, quizzes and learner performance.

4. Product Scope and Platform Strategy

The first product is a responsive web application. Students and teachers use a browser URL on Android, iOS, Windows, macOS and supported tablets. A Progressive Web App may provide an installable shortcut and supported notifications, but browser and operating-system restrictions still apply.

Client	Release	Notes
Desktop/laptop browser	MVP	Full teacher, student and admin workflows
Android browser/PWA	MVP	Responsive student and teacher experience; validate camera/mic and background behaviour
iPhone/iPad Safari/PWA	MVP	Responsive experience subject to iOS browser limitations
Native Android	Later	Only if PWA limitations or customer demand justify it

Client	Release	Notes
Native iOS	Later	Only if PWA limitations or customer demand justify it
Desktop app	Later	Optional explicit app-focus features with user notice and permission

5. Functional Requirements

Priority definitions: P0 = required for MVP; P1 = important follow-up; P2 = later enhancement.

5.1 Authentication and identity

- **AUTH-01 (P0):** Users can sign in securely using institution-approved credentials.
- **AUTH-02 (P0):** Each user has one or more assigned roles and institution membership.
- **AUTH-03 (P0):** Private class access requires a valid authenticated session and enrollment/permission.
- **AUTH-04 (P0):** Password reset, session expiry and sign-out are supported.
- **AUTH-05 (P1):** Support Google sign-in and email OTP where configured.
- **AUTH-06 (P1):** Optional identity re-verification for higher-risk sessions, with an accessible alternative.

5.2 Institution and tenant management

- **TEN-01 (P0):** Super Admin can create, suspend and configure an institution.
- **TEN-02 (P0):** Institution Admin can manage teachers, students, groups, courses and class rosters.
- **TEN-03 (P0):** Every institution-owned record is scoped to a tenant.
- **TEN-04 (P1):** Bulk import users from a validated CSV template.
- **TEN-05 (P1):** Configure academic year, timezone, attendance policy and retention settings.

5.3 Live virtual classroom

- **ROOM-01 (P0):** Teacher can create, schedule, start and end a classroom session.
- **ROOM-02 (P0):** Authorized students can join through the timetable or a protected invitation.
- **ROOM-03 (P0):** Provide microphone, camera, speaker and device selection controls where supported.
- **ROOM-04 (P0):** Provide participant list, chat, hand raise and teacher mute/remove controls.
- **ROOM-05 (P0):** Teacher can enable a waiting room and admit/reject waiting participants.
- **ROOM-06 (P0):** Teacher can lock a room and restrict participant screen sharing.
- **ROOM-07 (P1):** Screen sharing, presentation/file sharing and collaborative whiteboard.
- **ROOM-08 (P2):** Breakout rooms and consent-based recording.

5.4 Attendance and session events

- **ATT-01 (P0):** Record join, leave, reconnect and session end timestamps.
- **ATT-02 (P0):** Calculate connected duration from event intervals, avoiding double-counting reconnects.
- **ATT-03 (P0):** Apply institution-configurable late-entry and attendance thresholds.
- **ATT-04 (P0):** Distinguish temporary network disconnects from explicit leave events where the media provider supports it.
- **ATT-05 (P0):** Teacher can review and correct a record with a reason and audit trail.

- **ATT-06 (P1):** Export attendance as CSV/PDF and view course-level trends.

5.5 Participation and engagement

- **ENG-01 (P0):** Teacher can launch polls and quick questions during a live class.
- **ENG-02 (P0):** Student responses are timestamped and linked to the class session.
- **ENG-03 (P0):** Teacher can view participation counts and response summaries.
- **ENG-04 (P1):** Schedule random or topic-triggered knowledge checks.
- **ENG-05 (P1):** Show explainable indicators such as response rate, hand raises and quiz completion.
- **ENG-06 (P2):** Optional camera-based presence or head-orientation cues only after privacy, bias and accuracy evaluation.

5.6 Classroom safety and moderation

- **SAFE-01 (P0):** Prevent non-enrolled users from entering private classes unless a teacher explicitly admits them.
- **SAFE-02 (P0):** Teacher can remove a participant and block re-entry for the current session.
- **SAFE-03 (P0):** Teacher can disable chat, mute participants and restrict screen sharing.
- **SAFE-04 (P0):** Record security-relevant events in an audit log.
- **SAFE-05 (P1):** Rate-limit repeated join attempts and chat spam.
- **SAFE-06 (P1):** Provide report/flag workflows and institution review.

5.7 Learning tools and AI assistant

- **LEARN-01 (P1):** Teacher can create MCQ quizzes and assignments.
- **LEARN-02 (P1):** Objective questions can be automatically scored; descriptive answers remain teacher-reviewed.
- **AI-01 (P2):** Generate a lecture summary from approved transcript content.
- **AI-02 (P2):** Generate revision questions and key-topic notes with teacher review.
- **AI-03 (P2):** AI outputs must be labelled, traceable to source material where possible and editable.

5.8 Dashboards and reporting

Dashboard	MVP contents
Student	Upcoming classes, join button, attendance history, quizzes/polls, resources
Teacher	Today's classes, live participant list, waiting room, attendance, participation summary
Institution Admin	Users, classes, teacher assignments, institution-wide attendance and audit logs
Principal/HOD	Aggregated class/course trends and downloadable reports
Parent/Guardian	Later: linked learner attendance and institution-approved progress

6. Key User Journeys

Journey A — Institution setup

Super Admin creates institution → Institution Admin receives secure access → Admin configures courses/classes → imports or creates teachers and students → assigns roles and rosters → schedules first class.

Journey B — Teacher starts a class

Teacher signs in → opens scheduled class → checks microphone/camera → starts session → students enter waiting room or are admitted automatically if policy permits → teacher moderates class → launches a poll/quiz → ends session → reviews attendance and participation.

Journey C — Student attends a class

Student signs in → sees timetable → selects class → system checks enrollment and session status → student joins → receives clear camera/microphone permission prompts → participates in class activities → leaves or is disconnected → attendance record is updated.

Journey D — Unauthorized join attempt

Unknown user opens a private class link → server rejects access or places the user in a controlled waiting state → teacher receives an admission request only if policy allows → attempt is logged → teacher rejects or admits the user.

7. Business Rules and Data Definitions

Attendance calculation

Attendance is derived from verified session connection intervals, not from a single join event. The institution configures the minimum connected-time percentage and late-entry policy. A teacher override must include a reason and be auditable. A disconnected student may rejoin; intervals are combined without overlap.

Engagement interpretation

The product must distinguish participation from attention. Poll answers, quiz completion, hand raises and chat contributions are observable participation events. Tab visibility or camera-derived cues are weak signals and must not independently determine attendance, grades or discipline.

Core entities (initial logical data model)

Entity	Purpose / key relationships
Institution	Tenant boundary; owns users, courses, classes and policies
User	Identity and account; can have institution memberships
Membership / Role	Maps user to institution and role/permissions
Course / Group	Academic grouping and roster context
Classroom	Reusable class definition with teacher and enrolled learners
ClassSession	One scheduled/live occurrence of a classroom
ParticipantSession	Per-user join/leave/reconnect intervals
AttendanceRecord	Computed result plus overrides and reason
EngagementEvent	Poll, quiz, hand raise and other participation events
SecurityEvent	Admission, rejection, removal, lock and moderation events

Entity	Purpose / key relationships
Assessment	Quiz/assignment definition and submissions
AuditLog	Who changed what, when and why

8. Non-Functional Requirements

Area	Requirement / initial target
Responsive design	Core flows usable at phone, tablet and desktop widths
Performance	Standard API p95 target under 500 ms under agreed test load; media quality adapts to network
Availability	Initial service objective 99.5%, excluding planned maintenance and third-party outages
Scalability	Architecture supports multiple institutions and concurrent sessions; capacity validated by load tests
Security	TLS in transit, encryption at rest where supported, secure secrets and least-privilege access
Tenant isolation	Tenant scoping enforced server-side and tested for cross-tenant access
Accessibility	Keyboard navigation, visible focus, captions/transcripts where available, readable contrast
Compatibility	Current stable Chrome, Edge, Firefox and Safari; mobile browser support tested separately
Observability	Structured logs, error monitoring, metrics and health checks without unnecessary personal data
Backup/recovery	Documented backup schedule and tested restore process before production

9. Privacy, Safety and Compliance Requirements

- Explain what data is collected, why it is collected, who can view it and how long it is retained.
- Request camera and microphone permissions through standard browser prompts; never activate them secretly.
- Make optional monitoring opt-in and visible. Provide an equivalent participation path for users who cannot or do not use a camera.
- Do not collect personal browsing history, keystrokes, messages from other apps or unrelated application content.
- Avoid emotion recognition and claims that face direction proves attention.
- Do not make high-impact decisions about a student solely from automated indicators.
- Use role-based access, audit logs, data minimization, retention limits and secure deletion.
- For school-age users, design guardian/consent and safeguarding workflows and obtain legal review for applicable Indian laws and institutional policies.
- Recording, transcription and AI processing require clear notice, appropriate authorization and configurable retention.

10. Technical Architecture Direction

Layer	Proposed technology	Reason
Web client	Next.js, React, TypeScript	Responsive web app and shared UI

Layer	Proposed technology	Reason
UI system	Tailwind CSS, shadcn/ui	Accessible reusable components
API backend	Python FastAPI	Typed API services and AI integration
Database	PostgreSQL	Relational tenant-aware data
Authentication	Supabase Auth or equivalent	Managed identity; authorization remains server-enforced
Realtime media	LiveKit	Managed/self-hostable WebRTC media infrastructure
Realtime app events	WebSockets / provider webhooks	Presence, polls, moderation and session events
Cache/queue	Redis (when needed)	Rate limits, transient state and background jobs
AI services	OpenAI API (optional modules)	Summaries and question generation
Computer vision	OpenCV / MediaPipe (later evaluation)	Only for consent-based, validated signals
Storage	S3-compatible object storage	Files and optional recordings
Deployment	Vercel + container hosting	Web frontend and API deployment

Architecture rules

- Keep business logic in the backend; never trust client-side role or attendance claims.
- Use institution_id/tenant context in every tenant-owned query and enforce it centrally.
- Use signed, short-lived media access tokens generated by the backend.
- Treat provider webhooks as untrusted until signature and replay checks pass.
- Separate media transport from classroom business records so providers can be changed later.
- Use background jobs for reports, exports and AI tasks rather than blocking live class APIs.

11. Open-Source Reference Projects

The following projects are research references, not pre-approved dependencies. Before copying or integrating code, inspect the repository, activity, dependencies, security posture, license and compatibility. Reuse only components whose licenses and architecture fit the product.

Project	Potential learning area	Reference
AI Classroom Monitor	Computer vision / classroom monitoring concepts	github.com/Yuga-04/AI-Classroom-Monitor
ProxyMukt Attendance System	Attendance and identity workflow ideas	github.com/Sumant3086/ProxyMukt-Attendance-System-
AI-Assisted Classroom Monitoring	Video analysis and monitoring concepts	github.com/reevon-dsouza/AI-Assisted-classroom-monitoring
LiveKit Meet	Video meeting application patterns	github.com/livekit/meet
LiveKit Components	Reusable real-time media UI components	github.com/livekit/components-js
Jitsi Meet	Open-source conferencing architecture	github.com/jitsi/jitsi-meet
BigBlueButton	Education-focused virtual classroom capabilities	github.com/bigbluebutton/bigbluebutton

Repository validation checklist

- Confirm the repository is accessible and inspect its current source rather than relying on its README alone.
- Read LICENSE and identify obligations for distribution, modification and attribution.
- Check last commit, open issues, dependency age, build instructions and test coverage.
- Run in an isolated environment; do not import unknown binaries or secrets.
- Create a feature-by-feature comparison before deciding whether to reuse, adapt or simply learn from it.

12. MVP Scope and Release Plan

Release	Included scope	Exit condition
MVP 0.1	Auth, tenant setup, roles, classes, responsive shell	Institution admin can create a class and roster
MVP 0.2	LiveKit classroom, waiting room, participant controls	Teacher and enrolled students complete a stable live session
MVP 0.3	Join/leave events, attendance calculation, audit log	Attendance report reconciles with session events
MVP 0.4	Polls, quick quizzes, participation dashboard	Teacher can run an activity and view results
MVP 0.5	Security hardening, mobile browser QA, exports	Pilot-ready with documented limitations and support process
Later	AI summaries, advanced analytics, parent view, native apps	Built only after user feedback and privacy review

13. MVP Acceptance Criteria

- An institution admin can create an institution workspace and invite/manage teachers and students.
- A teacher can create and schedule a class and assign an enrolled roster.
- An enrolled student can join from a supported laptop and mobile browser.
- A non-enrolled user cannot bypass authentication/enrollment checks using a copied class URL.
- Teacher can admit/reject waiting participants, mute/remove a participant and end the session.
- The system records join, leave and reconnect events and calculates connected duration without double-counting.
- Teacher can launch a poll and view student responses.
- Tenant A users cannot access Tenant B records through API or UI; automated tests cover this.
- Camera/microphone use is permission-based and clearly indicated.
- Core pages pass responsive checks at representative phone, tablet and desktop widths.
- Errors are logged without exposing passwords, access tokens or unnecessary sensitive student data.

14. Success Metrics

Metric	Definition
Successful class start rate	Percentage of scheduled sessions that start successfully
Join success rate	Successful authorized joins divided by join attempts
Attendance record completeness	Sessions with complete event-based attendance data
Participation rate	Students completing at least one defined interaction

Metric	Definition
Moderation event rate	Security/moderation events per class hour; interpret with context
Teacher task completion	Teachers able to start class and retrieve report without support
Reliability	Session/API uptime and error rate
Mobile usability	Join and participation task success on supported mobile browsers

Do not optimize for

A single opaque attention score, number of camera checks, or number of flags. These can incentivize invasive monitoring and create misleading conclusions.

15. Risks, Dependencies and Mitigations

Risk	Impact	Mitigation
Mobile browser limitations	Camera/mic, background and notification behaviour varies	Test on real devices; document limits; consider native apps only when need
Network instability	Dropouts and disputed attendance	Reconnect handling, event intervals, teacher review and network-aware me
False monitoring signals	Unfair flags or loss of trust	Use observable participation; label weak signals; no automated penalties
Minor/student privacy	Legal and reputational risk	Data minimization, guardian workflows, consent, retention and legal review
Video infrastructure cost	Unexpected operating expense	Usage limits, pilot sizing, monitoring and provider cost review
Open-source license conflict	Distribution restrictions	License audit before reuse; isolate or replace incompatible components
Tenant data leakage	Critical security failure	Server-side tenant enforcement, automated isolation tests and security revi

16. Decisions Required Before Production

- Final product name and domain availability.
- Initial pilot institution and expected concurrent classroom size.
- Whether video infrastructure is LiveKit Cloud or self-hosted for the pilot.
- Institution-specific attendance thresholds and correction policy.
- Data retention periods for attendance, chat, recordings and transcripts.
- Applicable consent and guardian workflows for each customer segment.
- Hosting region, backup and incident response requirements.
- Subscription packaging and pricing after pilot feedback.

17. Immediate Build Sequence

Step	Deliverable
1	Create repository and project folder structure
2	Write architecture decision records and environment setup guide

Step	Deliverable
3	Design database schema and tenant isolation model
4	Build responsive design system and role-based app shell
5	Implement authentication and institution membership
6	Implement class/session management
7	Integrate LiveKit and secure room-token generation
8	Implement attendance event pipeline and reports
9	Implement polls, quizzes and participation reporting
10	Run security, accessibility, mobile and load testing before pilot

18. Glossary

Term	Meaning
Tenant	An institution workspace whose data is isolated from other institutions
PWA	Progressive Web App; a web application that can be installable on supported devices
WebRTC	Browser/mobile technology for real-time audio and video communication
Attendance interval	A verified period during which a participant is connected to a session
Participation event	An observable interaction such as a poll response, quiz answer or hand raise
Engagement indicator	A limited signal that may help a teacher; not proof of attention
RBAC	Role-based access control
MVP	Minimum Viable Product; the smallest useful release for pilot validation

End of PRD — Version 1.0